Title of Paper: Software Engineering

Sr.No	Heading	Particulars
1	**Description the course: Including but Not limited to:**	This course provides an in-depth understanding of Scrum, an Agile framework for developing, delivering, and sustaining complex products. Students will learn the principles and practices of Scrum, including roles (Scrum Master, Product Owner, Development Team), events (Sprint, Scrum Meetings), and artifacts (Product Backlog, Sprint Backlog, Increment). The course emphasizes hands-on exercises, real-world scenarios, and collaborative activities to master iterative development and enhance team productivity. By the end, learners will be equipped to implement Scrum in software engineering projects effectively and drive organizational agility.
2	**Vertical :**	Major
3	**Type :**	Theory
4	**Credits :**	2 credits (1 credit = 15 Hours for Theory in a semester, Total 30 hours)
5	**Hours Allotted :**	30
6	**Marks Allotted:**	50
7	**Course Objectives (CO):** **CO1:** Understand the core principles of Agile and the Scrum framework. **CO2:** Explore the high-level Scrum process and its key components. **CO3:** Develop skills in managing the Product Backlog effectively. **CO4:** Learn techniques for Sprint planning, execution, and tracking. **CO5:** Gain insights into Scrum-based project, quality, and risk management. **CO6:** Master the art of writing clear and actionable user stories.	
8	**Course Outcomes (OC):** **OC1:** Demonstrate a comprehensive understanding of Agile concepts and Scrum practices. **OC2:** Apply Scrum processes to effectively manage software development life cycles. **OC3:** Create and prioritize user stories for efficient Product Backlog management. **OC4:** Utilize metrics to evaluate and enhance Sprint performance and team productivity. **OC5:** Implement strategies for cost, customer, and risk management in Scrum projects. **OC6:** Formulate effective Sprint retrospectives to drive continuous improvement.	
9	**Module 1:**	
	Software and Software Engineering, Process Models, Introduction to Agile Concepts, All about Scrum, Scrum Process: High-Level View. Product Backlog Management, Sprint Planning, Writing Effective User Stories, Sprint Execution and Tracking, Sprint Review, Sprint Retrospectives	15 Hrs
	Module 2:	
	Measurements and Metrics in Scrum, Software Development Life Cycle and Waterfall Model, Project Management in Scrum and Waterfall, Quality Management in Scrum, Customer Management in Scrum, Risk Management in Scrum, Cost Management in Scrum.	15 Hrs

<table>
<tr><td>10</td><td colspan="2">Books and References:
1. “Agile Scrum”, Rama Bedarkar, Wiley, 1st, 2020
2. "Mastering Professional Scrum: A Practitioner's Guide to Overcoming Challenges and Maximizing the Benefits of Agility" by Stephanie Ockerman and Simon Reindl, Addison-Wesley Professional, 1st edition (2019).
3. "Scrum: A Pocket Guide" by Gunther Verheyen, Van Haren Publishing, 2nd edition (2019).
4. "Software in 30 Days" by Ken Schwaber and Jeff Sutherland, Wiley, 1st edition (2012).
5. "Scrum Insights for Practitioners: The Scrum Guide Companion" by Hiren Doshi, PracticeAgile Solutions, 1st edition (2016).
6. "A Scrum Book: The Spirit of the Game" by Jeff Sutherland and James O. Coplien, Pragmatic Bookshelf, 1st edition (2019).
7. "The Scrum Fieldbook: A Master Class on Accelerating Performance, Getting Results, and Defining the Future" by J.J. Sutherland, Random House Business, 1st edition (2019).</td></tr>
<tr><td>12</td><td>Internal Continuous Assessment: 40%</td><td>Semester End Examination: 60%</td></tr>
<tr><td>13</td><td>Continuous Evaluation through:
Class test of 1 of 15 marks
Class test of 2 of 15 marks
Average of the two: 15 marks
Quizzes/ Presentations/ Assignments: 5 marks
Total: 20 marks</td><td>Format of Question Paper: External Examination (30 Marks)– 1 hr duration</td></tr>
<tr><td>14</td><td colspan="2">Format of Question Paper: (Semester End Examination: 30 Marks. Duration:1 hour)
Q1: Attempt any two (out of four) from Module 1 (15 marks)
Q2: Attempt any two (out of four) from Module 2 (15 marks)
Or
Q1: Attempt any three (out of five) from Module 1 (15 marks)
Q2: Attempt any three (out of five) from Module 2 (15 marks)</td></tr>
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